CSE 6242 Final Report - Team 40

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1 Introduction

Within the video game industry, studios of different sizes optimize for different commercial outcomes [1]. AAA studios (that employ development teams numbering in the dozens to hundreds) often prioritize unit sales and revenue maximization, while small studios (that employ teams of 15 or fewer in total) value player engagement and community sentiment alongside traditional revenue metrics. Regardless of which outcomes a studio prioritizes, the path to achieving business objectives increasingly runs through the interpretation of market trends and data, including pricing behavior, player engagement, competitive positioning, and player feedback. AAA studios commonly have dedicated analytics, insights, and market intelligence teams to translate raw data into pricing, game development, and marketing decisions. In contrast, independent (indie) developers and small studios rarely employ a single data analyst, and yet are making the same types of decisions without the specialized personnel to validate them with analyses.

Steam, the largest PC digital distribution platform, publishes extensive data on pricing, reviews, tags, genres, release dates, and more. While, theoretically, indie developers have the same potential to use the same or similar market data as their AAA counterparts, in reality, this data is only accessible through fragmented public API and ad-hoc scrapers. Across the full catalog, Steam data is too large for repeated manual ingestion, and converting raw data into interpretable outputs requires data engineering, statistical modeling, and data visualization expertise that indie developers rarely have in-house.

To address the existing deficiency in available analytics tools geared towards indie developers, we built an interactive Tableau dashboard that translates Steam data into actionable insights and visualizations. By emphasizing usability and surfacing relevant metrics for indie developers—revenue predictions, genre trends, review-based engagement indicators, and pricing benchmarks, our goal is to democratize access to market intelligence and support developers with little to no data science background.

2 Problem Definition

Indie developers face three major challenges when using Steam data. First, they often lack analytics-fluent personnel. Second, existing Steam analytics tools typically require, at the very least, familiarity with Python, SQL, or other statistical expertise and only focus on a subset of data (e.g. sentiment analysis of reviews only) as opposed to the multiple dimensions involved in development and distribution decisions. Third, Steam provides raw data inputs but no intuitive, front-end layer that non-technical users can interact with directly.

Our dashboard is designed to answer four analytical questions an indie developer faces during development and/or at launch: (1) What commercial outcomes can we expect? (2) How do our metrics compare against industry benchmarks? (3) What game features are most influential on player engagement? (4) Can we quickly obtain these answers without dedicated analytics staff? We attempt to address these concerns by combining multiple Steam data sources into a single application.

Given n number of datasets $D = \{d_1, d_2, d_3, ..., d_n\}$, let G be a finite set of games where $g_i \in G$. Each game g_i is defined as a set of variables represented by a vector x_i. We intend to use a function $f(\mathrm{x}_i) = \hat{y}$, where R_i is the number of reviews observed for game g_i to obtain $\hat{y} = \log(R_i + 1)$. The objective is to minimize the prediction error between the estimated outcomes $\hat{y}$ and the observed values in G for each game g_i.

3 Literature Survey

To identify factors that are correlated with various commercial outcomes, we reviewed a variety of related literature and identified four frequently-cited metrics: pricing, visibility, player engagement, and player feedback.

Recent studies have applied predictive modeling and machine learning to map factors to game performance, training regression or classification models on features at launch time (price, genre tags, release timing, early review counts) to predict total sales, review count, or player retention [2], [3], [4], [5], [6], [7]. Additionally, broader reviews highlight how behavioral, economic, and platform-specific data are being integrated into development and release strategies, emphasizing the growing importance of analytics in guiding business decisions [8]. Steam has emerged as a dominant source of market data for research and practical analysis. Prior work has included analyzing review sentiment and length across popular game genres [9], tracking dynamic genre trends through tagging behavior [10], and examining publishing patterns on the platform [11]. Additional studies have focused on technical methods for collecting and visualizing Steam data, demonstrating the potential for actionable insights from these publicly available resources [12], [13].

The disparity in analytical capacity between AAA and indie developers is documented in research: AAA studios maintain in-house analytics teams to inform decisions on release planning, pricing, and portfolio management, while indie teams lack that personnel [1], [14]. The issue then, for indie developers, largely points to a skill gap rather than a data gap.

Interactive dashboards are widely recognized as effective tools for bridging this gap, transforming complex datasets into intuitive visual summaries that non-technical users can interpret directly [15]. Research on dashboard design for non-experts emphasize usability, guided interaction, and intelligent visualization design as critical factors [16], [17]. Within gaming, dashboards and visualization tools have been used to forecast industry trends, identify KPIs, and explore Steam data directly [13], [18], while broader studies show that visual analytics can enhance competitive decision-making [19].

From our review of the literature, we've concluded that no tool exists that integrates publicly available Steam data into a practical dashboard tailored indie developers, presenting revenue estimates, genre trends, review behavior, and player engagement in a single interface without needing analytics expertise.

4 Proposed Method

We developed a user-friendly dashboard that transforms raw Steam data into actionable insights for indie developers. Unlike prior approaches that focus on isolated analytics tasks or require advanced technical expertise, our method integrates interpretable modeling with an interactive interface, enabling real-time scenario analysis for non-technical users. The method follows a structured workflow that includes data collection, preprocessing, algorithm development, and dashboard design (Figure 1), with the goal of maximizing usability while providing reliable market intelligence through descriptive and predictive insights.

Our team retrieved data from three primary sources: (1) the official Steamworks Web API, (2) internal Steam JSON endpoints, and (3) raw HTML content from the Steam website, resulting in more than +200,000 App IDs across 11 initial datasets. These datasets were gathered using the Python Requests library to connect with the API and JSON endpoints, parsed HTML pages using BeautifulSoup, ensured schema validation across datasets using Pydantic, and performed data manipulation and CSV serialization using Pandas.

After filtering the dataset d_1 for the App ID type of 'game', we performed left joins from 6 other datasets to d_1, creating the 'Master' data table for our regression model of size $(m \times n)$. Within these 6 datasets, we took the top n unique categorical data points by value count in descending order, to reduce overfitting in our models. These categorical variables were then one-hot encoded and aggregated back into one line per game, resulting in a final dataset of size (289,043 x 108). Our aggregated data schema is shown in Figure 2.

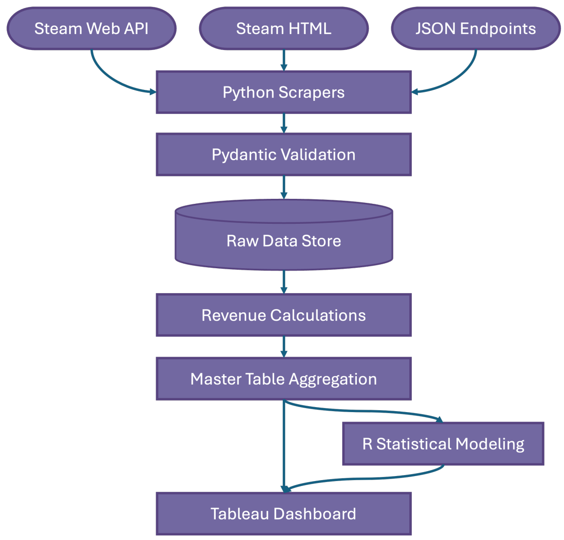

Figure 1: End-to-end pipeline from raw data to an interactive decision support tool.

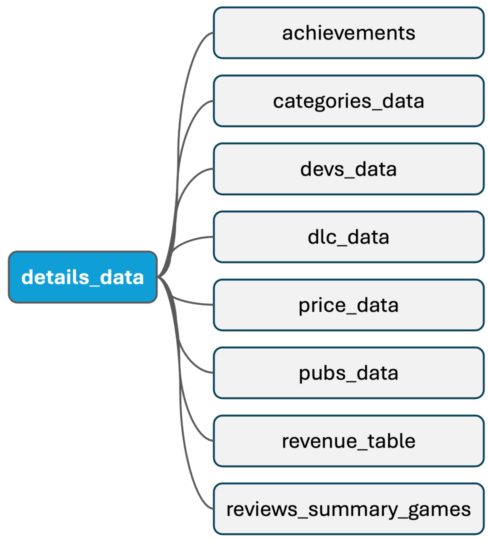

Figure 2: Data schema of Master dataset, created by joining tables on the App ID 'game'.

We encountered a challenge with the estimation of Units Sold (U) and Revenue (R) in our Regression Model, given the lack of publicly available data. We overcame this using the Boxleiter Method with a slight modification of adding bound limits, where i represents the current observed game, M is a Scaling Multiplier and B is the Bound Limit Adjustment. Our dashboard calculates these metrics as $U = \text{Reviews}_i \times (M \pm B)$ and $R = U \times \text{Price}_i$. We chose to normalize the reviews data in our model using $\log(\text{Total Reviews}+1)$ for our dependent variable to minimize variance within the model for outliers in the training dataset.

Finally, our team built the tool using Tableau, focusing on interactivity for predictions of three distinct areas, each contained within a dedicated tab: Sales, Exploratory Data Analysis (EDA), and Recommendation. The dashboard outputs are based on the parameters set by the user. We are structuring this to be a dynamic decision-support tool rather than a static report.

Sales: The Sales tab uses the predicted review volume in conjunction with the modified Boxleiter multiplier to estimate Units Sold and Revenue. We implemented Tableau parameters and sets that act as interactive inputs, allowing users to toggle Game Metadata (e.g., 'Action', 'Multiplayer', 'Early Access'), adjust a Price Slider, Achievements Count Sliders, DLC Count Slider, and Review Score Slider. These inputs are tied to Calculated Fields that leverage our regression coefficients to provide real-time Units Sold and Revenue predictions. This was then plotted against similar games based on these selections to give the user a sense of where they compare to the market and gather insights into their competition.

Exploratory Data Analysis (EDA): The EDA tab provides visualizations for identifying outliers, top genres by engagement, and genre pricing strategies and their effects on engagement.

These also include interactive features to drill-down further into the different Genres, Pricing Strategies, and Specific Games to compare the effectiveness of certain features.

Recommendations: The Recommendations tab provides a baseline against all games within our dataset, and given the parameters set by the user, it will provide an action to optimize their game to compete with similar games from the selections.

The tool's innovation lies in the domain-specific translation of raw data into an actionable decision support system. We optimized the interface for low-latency cognitive processing, allowing stakeholders to perform complex 'What-If' scenarios in real-time across all features, rather than a subset. The novelty of our tool is the bidirectional interaction model. By integrating Tableau's filtering logic with our regression outputs, we enable iterative discovery, where a user can dynamically isolate variables to observe the model's sensitivity, a feature often missing in black-box ML systems. Although these 'black-box' models offer marginal accuracy gains, they lack the explainability required for business decisions and directional insights. We intentionally utilized a parsimonious linear regression model to ensure that the UI remains transparent and trustworthy, providing users with clear coefficients rather than obscure predictions.

5 Evaluation

To assess the effectiveness of the dashboard, we applied statistical analyses to the underlying algorithm to evaluate (1) the predictive validity of the model, and (2) the utility of the dashboard for non-technical users, using the analytics questions outlined in the Problem Definition section as our guides.

We first trained our linear regression model on log(Total Reviews + 1), as described in the proposed method. Our original model resulted in a Residual Standard Error (RSE) of 0.9011. Since the model uses a logarithmic scale, this error corresponds to a multiplicative factor of approximately 2.46, meaning that predicted review counts are typically within a factor of 2.46 of the actual total Reviews. The model achieved an R^2 value of 83.65%, indicating that a substantial portion of the variance in review counts is explained by the selected variables. The Adjusted R^2 was identical, confirming that each variable contributes meaningful predictive power without introducing unnecessary complexity or noise.

Table 1: Top 15 variables by VIF score

Variable_Name	VIF
cats_description_Full.controller.support	203.22
controller_support_full	203.03
cats_description_PvP	11.76
cats_description_Co.op	11.58
cats_description_Online.PvP	7.20
cats_description_Multi.player	7.18
cats_description_Online.Co.op	7.08
tag_name_Early.Access	7.04
genres_description_Early.Access	7.02
cats_description_Shared.Split.Screen	6.63
cats_description_Shared.Split.Screen.PvP	4.88
genres_description_RPG	4.83
tag_name_RPG	4.76
tag_name_Strategy	4.74
genres_description_Strategy	4.74

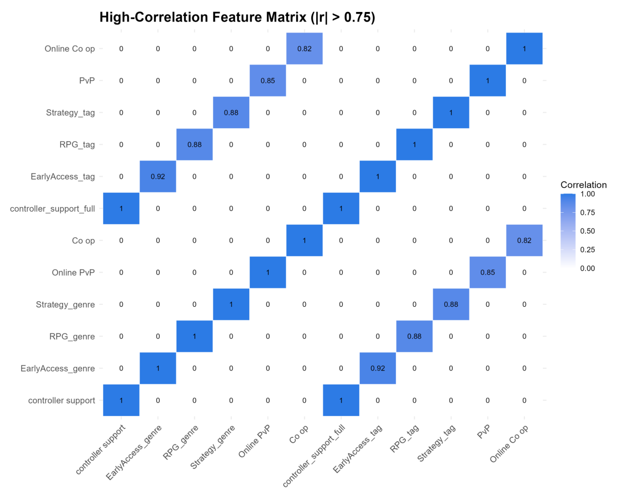

Figure 3: High-correlation feature matrix ($|r| > 0.75$)

To further validate the robustness of the model, we analyzed the multicollinearity among the predictor variables using the Variance Inflation Factor (VIF) score. Table 1 shows the top 15 VIF score variables in descending order, which were then used to create a correlation matrix. This analysis revealed pairs of variables with correlations exceeding 75% and can be seen in Figure 3. To reduce redundancy, we removed the first column of these pairs (Variable_1) and retrained the model on the trimmed dataset of size (201367 x 93). After retraining, we observed marginal negative shifts in RSE to 0.9021 and a slight decrease in R^2 to 83.61%. These minimal differences indicate that the removed variables had little impact on predictive performance despite their high multicollinearity, suggesting that the model is robust to feature redundancy. The decision to remove these variables was therefore driven by the desire to improve interpretability and reduce duplication, as many of the highly correlated features represented overlapping concepts across Genres, Tags, and Categories.

From a system perspective, the dashboard enables rapid generation of insights through interactive "what-if" analysis, allowing users to adjust parameters and immediately observe changes in predicted outcomes. This significantly reduces the time required to explore potential strategies compared to traditional static analysis workflows. Furthermore, by benchmarking user-defined configurations against similar games, the dashboard provides meaningful context for performance and supports competitive analysis. Integration of regression output with exploratory visualizations also highlights which features, such as genre, pricing, and engagement-related attributes, have the greatest impact on success, allowing more informed design and business decisions.

6 Conclusions and Discussions

This project produced a working dashboard that allows independent developers and small game studios to generate predictions from Steam data, explore market trends, and benchmark themselves against comparable titles. Several limitations emerged across the project's three major components (data collection, model performance, and visualization) which we discuss below, along with directions for future work.

Data Collection: Without funding for a shared database service, we maintained our dataset as locally stored CSV files circulated among team members via SharePoint. Migrating this workflow to a self-hosted or cloud-based database, with scripts versioned in our GitHub repository, would improve replicability for future iterations. Additionally, we did not have any temporal aspects to our dataset given the time constraints of the project and limitations of our machines, which could be useful for gathering time dimensionality for current player bases. To avoid getting a 429 'Rate Limit Exceeded' error, we chose to build in rate limiting in our Python scripts of 1.5 seconds between each request; given that each scraped section of the Steam catalog contained +200,000 App IDs, processing each section (if not parsable within the same request) ranged from 3.5-7 days, accounting for time to parse, validate, and append to the current CSV. We ran the scripts on two machines, and without having a way for us to simultaneously use multiple IP Addresses across each script, we were limited to pulling the entire dataset in just over a month.

Model Performance: The largest coefficients from our Linear Regression model were those of Review Scores, specifically those at either end of the range (ratings of 9 and 1), while other variables not related to Review Scores were all below 1. Although many other coefficients were statistically significant, they led to little or no improvement in Units Sold across games. Given our methodology for predicting Units Sold (and subsequently Revenue) removing the noise associated with Review Score variables from the model could give developers better insight into features that drive sales or engagement. One unexpected insight: the model suggests that you can make a lot of money regardless of whether the game is terribly rated or extremely well.

Visualization: While Tableau is approachable for non-technical users, the limited project runway—in addition to the other course assignments, full-time jobs, and personal obligations—constrained how much we could improve on the dashboard's design. Further iteration could surface important signals more prominently than in the existing tool.

Future Work: The next steps of this research would be to incorporate user testing of the target audience. Conducting experiments on application testing to get feedback from game developers and other industry experts would help us to validate which predictions and features are most useful in practice as well as identify parts of the dashboard that may be confusing.

Team Contributions and Project Timeline: The work for this project was distributed evenly among group members and labeled under three categories: data collection, algorithm implementation, and visualization. Outlined in Figure 4, our revised plan was structured into three major milestones with deliverables: Proposal, Progress Report, and Final Report. Meanwhile, the aforementioned areas related to developing the dashboard are broken into two separate parts to allow for a pivot if needed during the progress reporting phase. This plan remained largely the same as our initial plan of activities stated in the Proposal (Figure 5), with few additional deadlines.

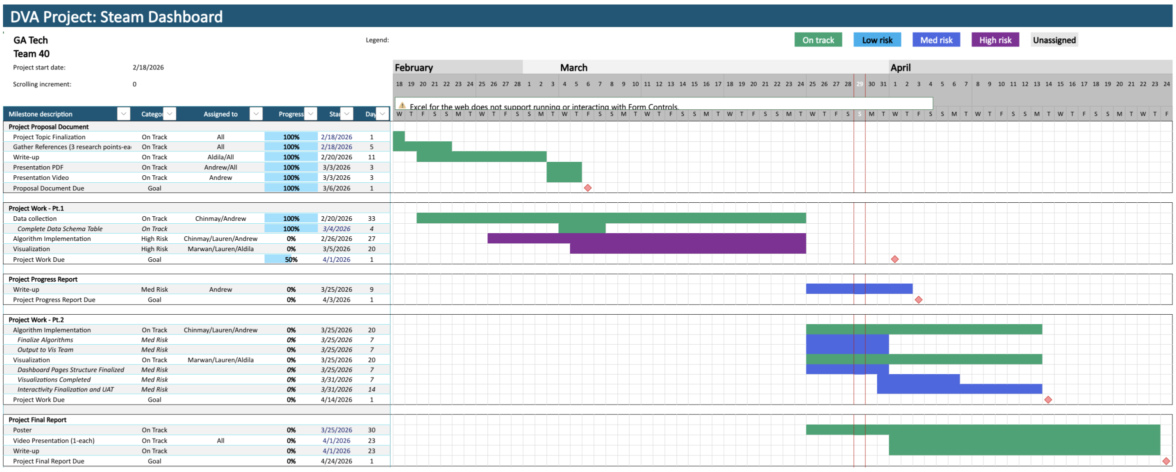

Figure 4: Team 40 Project Gantt Chart - Revised Plan

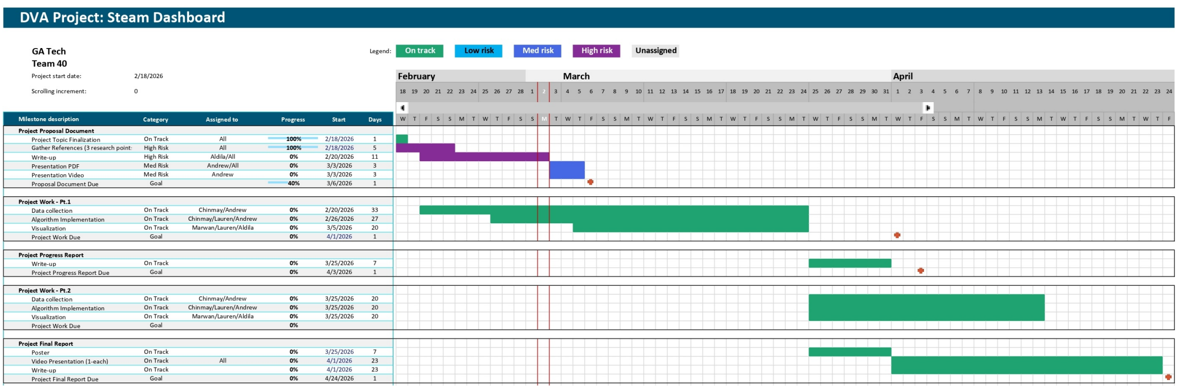

Figure 5: Team 40 Project Gantt Chart - Original Plan

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